

HISTORY — THE 9 NAMES EXAMINERS ASK

WORKER	YEAR	CONTRIBUTION — THE EXACT PHRASE TO WRITE
Lavoisier	1780	Father of Nutrition Science. Respiration = slow combustion; first animal calorimeter with Laplace
Liebig	1840s	Father of Agricultural Chemistry; proposed protein as sole source of muscular energy (<i>later disproved</i>)
Boussingault	1839	First N-balance studies in farm animals; field feeding trials
Henneberg & Stohmann	1860	Weende (Proximate) System , Weende Expt. Station, Germany
Atwater & Rosa	1897	First human respiration calorimeter; physiological fuel values 4:9:4
Kellner	1905	Starch Equivalent system (Germany)
Armsby	1915	Net Energy / Therm system (USA)
Hopkins → Funk	1906+12	"Accessory food factors" → coined "vitamine"
Van Soest	1963	Detergent fibre system (NDF/ADF/ADL) — cell wall vs cell contents

BODY VS PLANT — THE CONTRAST Q

FRACTION	ANIMAL	PLANT
Water	50–60%	5–90%
Protein	15–20%	~10%
Lipid	15–25%	2–3%
Carbohydrate	<1%	~75%
Ash	3–5%	5–10%

Write these 4 differences: ① Animal stores energy as **fat + glycogen**; plant as **starch + fructans**. ② Plant has a **cell wall** (cellulose, hemicellulose, lignin); animal has none. ③ Animal sterol = **cholesterol**; plant = **phytosterol**. ④ Animal N is almost all **true protein**; plant carries substantial **NPN** (nitrate, amides, alkaloids).
Age effect: newborn = water **75–80%**, fat **~2%**. As age ↑ → **water ↓**, **fat ↑**, protein & ash nearly constant. *Fat is the most variable body constituent; water the most abundant.*

NUTRITIONAL TERMS — 2-MARK DEFINITIONS

Feed/Feedstuff — any edible material furnishing nutrients. **Ration** — total feed allowed to one animal in **24 h**. **Diet** — feed(s) offered, without a time limit. **Balanced ration** — a ration supplying all nutrients in required **proportion & amount** for 24 h to maintain the animal in a specified state of production. **Maintenance ration** — supports life processes with **zero gain, zero loss, zero production**. **Production ration** — over and above maintenance.
Nutrient — a chemical substance that supports life. **Dry matter (DM)** — feed minus moisture (105 °C). **Concentrate** — **CF < 18% DM**, high TDN. **Roughage** — **CF > 18% DM**. **Supplement** — corrects a specific nutrient deficit (protein/mineral). **Additive** — non-nutritive, added to improve efficiency/quality. **Palatability** — feed characters that attract the animal & determine voluntary intake. **Ad libitum** — offered free-choice. **TMR** — roughage + concentrate mixed so animal cannot select. **Provender** — ground/crushed dry feed mixture.

WEENDE PROXIMATE ANALYSIS — THE FLOW TO DRAW

FEED 100 g	105 °C →	MOISTURE Loss on drying	600 °C →	TOTAL ASH residue	→	ORGANIC MATTER CP·EE·CF·NFE
FRACTION	METHOD / REAGENT			KEY NUMBER & TRAP		
Moisture / DM	Hot air oven 100–105 °C, 12–24 h			DM = 100 - moisture		
Total Ash	Muffle furnace 550–600 °C , 4–6 h			AIA (boil in 10% HCl) > 3% = sand adulteration		
Crude Protein	Kjeldahl: digest (H ₂ SO ₄ + K ₂ SO ₄ + CuSO ₄) → distil (NaOH) → titrate			CP = N × 6.25		
Ether Extract	Soxhlet, petroleum/diethyl ether, 16 siphons			Includes waxes, pigments, sterols → "crude" fat		
Crude Fibre	Boil 1.25% H ₂ SO ₄ then 1.25% NaOH, 30 min each			Loses up to 80% lignin + 50% hemicellulose into NFE		
NFE	By difference — never measured			NFE = 100 - (M+A+CP+EE+CF)		

N-factors: general **6.25** (16% N) · milk **6.38** · wheat **5.83** · rice **5.95** · groundnut **5.46**.

Chief defect of Weende: the CF/NFE split is **artificial** — lignin (the truly indigestible part) migrates into NFE, so NFE is often *less* digestible than CF in roughages. This single line answers "limitations of proximate analysis".

VAN SOEST DETERGENT SYSTEM — THE FIX

FEED		Neutral detergent	NDS (soluble) sugar, starch, protein, lipid, pectin
FRACTION	REAGENT	CONTAINS	
NDF	Neutral detergent (Na lauryl sulphate, pH 7)	Cellulose + Hemicellulose + Lignin + silica = total cell wall	
ADF	Acid detergent (CTAB in 1 N H ₂ SO ₄)	Cellulose + Lignin + silica	
ADL	72% H ₂ SO ₄ or KMnO ₄ on ADF	Lignin + silica	

Hemicellulose = NDF - ADF · Cellulose = ADF - ADL
NDF ⇒ predicts intake (rumen fill). ADF ⇒ predicts digestibility & energy (lignin). Both ↑ with maturity of the fodder ⇒ both intake and digestibility fall. **Cell contents ≈ 98% digestible; cell wall is variably digestible depending on lignification.**
NDF = Intake, ADF = Digestibility. "N for Nibble, A for Absorb."

WHY NUTRIENTS MATTER — PRODUCTION & HEALTH (THE OPENING PARAGRAPH FOR ANY LONG ANSWER)

Six nutrient classes: water, carbohydrate, protein, lipid, minerals, vitamins. **Functions grouped in 3:** ① **Energy-yielding** (CHO, lipid, protein) ② **Body-building / structural** (protein, minerals, water, lipid) ③ **Regulatory** (vitamins, minerals, water, enzymes/hormones). **Feed = 60–70% of total cost of livestock production** — hence nutrition is the single largest determinant of profitability. Under-nutrition ⇒ delayed puberty, anoestrus, low milk yield, low immunity (T-cell & antibody suppression), high neonatal mortality; over-nutrition ⇒ fatty liver, dystocia, ketosis, acidosis, lameness. **Nutrition–immunity–reproduction–production is one continuum** — that sentence earns marks in any "importance of nutrients" question.

CARBOHYDRATE NUTRITION

Classification: **Monosaccharides** (glucose, fructose, galactose, ribose, xylose, arabinose) → **Disaccharides** (sucrose = glu+fru; lactose = glu+gal, *only animal sugar*; maltose = glu+glu; cellobiose) → **Oligosaccharides** (raffinose, stachyose — flatulence factors of soybean) → **Polysaccharides:** **storage** = starch (amylose+amylopectin), glycogen, fructans, inulin; **structural** = cellulose (β-1,4), hemicellulose, pectin, lignin (*not a CHO — a phenolic polymer*).

Structural (fibrous) vs Non-structural (NSC): the key nutritional divide. Ruminants extract structural CHO via **microbial cellulase**; monogastrics cannot (no β-1,4 glycosidase) and depend on hind-gut fermentation only marginally (horse/rabbit are the exceptions — large caecum).

WoLin: 57.5 C₆H₁₂O₆ → 65 Acetate + 20 Propionate + 15 Butyrate + 60 CO₂ + 35 CH₄ + 25 H₂O

VFA	ROUGHAGE DIET	CONCENTRATE DIET	METABOLIC FATE
Acetate C2	65–70%	↓ 45–55%	Milk fat precursor; body fat; ketogenic
Propionate C3	15–20%	↑ 30–40%	Only glucogenic VFA → gluconeogenesis → lactose & blood glucose
Butyrate C4	10–15%	~15%	Rumen epithelium fuel; → β-hydroxybutyrate (ketogenic)

Normal acetate:propionate ≈ 3:1. Feeding ↑ concentrate narrows it → **milk fat depression** + ↑ body weight gain. VFAs supply **70–80%** of the ruminant's energy; methane wastes **6–8% of GE**.

Field line: ruminants are functionally **glucose-poor** — almost no dietary glucose is absorbed; blood glucose **40–60 mg/dL** (vs 80–120 in monogastrics) and is made from propionate + glucogenic AA + glycerol.

PROTEIN NUTRITION

Essential AA (10) — mnemonic:

P V T T I M H A L L — Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine. Poultry additionally need **Glycine/Serine**; the **cat** uniquely requires dietary **Taurine** and cannot spare Arginine.

Semi-essential: Cysteine spares Methionine; Tyrosine spares Phenylalanine. **Limiting AA:** **Lysine** in cereal/maize-based swine diets; **Methionine** in maize-soya poultry diets; **Lysine + Methionine** for milk protein in high yielders (hence rumen-protected Met/Lys).

Ruminant protein metabolism — the whole story in one line: dietary CP splits into **RDP** (degraded to peptides → AA → NH₃) and **RUP/bypass** (escapes to abomasum). Rumen microbes use NH₃ + carbon skeleton + ATP from fermentable energy to build **Microbial Crude Protein (MCP)**, which alone can meet **50–80%** of the animal's protein need and is of high, **constant biological value (~66–80)**. Excess NH₃ → absorbed → liver **urea cycle** → urine or **recycled via saliva & rumen wall** (*ruminohepatic N recycling* — the ruminant's N-conservation mechanism on low-protein straw diets).

Protein equivalent of urea = 45–46.6% N × 6.25 = 281–291% CP

Rumen NH₃-N optimum ≈ 5 mg/100 mL rumen fluid for maximum microbial growth. **N : S ratio 10–14 : 1** is mandatory when feeding NPN (S needed for Met, Cys, biotin, thiamine).

Protein quality is a monogastric concept. In ruminants, microbial fermentation levels out AA quality, so BV of the diet matters far less than **degradability (RDP:RUP)** and total N supply. Write this sentence for "measures of protein quality in ruminants vs non-ruminants".

LIPID NUTRITION

Classification: **Simple** (fats/oils = triglycerides; waxes) · **Compound** (phospho-, glyco-, lipoproteins) · **Derived** (fatty acids, glycerol, sterols, fat-soluble vitamins).

Functions: most concentrated energy source (**2.25× CHO**), carrier of vitamins A D E K, source of EFA, structural (cell membrane phospholipids), insulation, precursor of eicosanoids & steroid hormones, improves palatability, reduces dust & ration heat increment.

Essential fatty acids: **Linoleic (C18:2, ω-6)** — truly essential in all species; **α-Linolenic (C18:3, ω-3)**; **Arachidonic (C20:4)** — essential only for the **cat** (no Δ6-desaturase). Deficiency ⇒ scaly dermatitis, poor growth, hair loss, impaired reproduction, ↓ egg size.

Rumen biohydrogenation: dietary unsaturated FA are lipolysed then **hydrogenated to stearic acid (C18:0)** by rumen bacteria → ruminant depot fat is **hard & saturated** and dietary PUFA hardly reach tissues. Intermediates include **trans-10**, **cis-12 CLA** (the milk-fat-depressing isomer).

Bypass / protected fat: free oil above **3–4% of DM** coats fibre and poisons cellulolytic bacteria ⇒ fibre digestion crashes. Solution: **calcium salts of long-chain fatty acids (Ca-LCFA) / prilled fat**, inert at rumen pH 6.5, dissociating in the acidic abomasum. Fed at **100–150 g/day** to high-yielding crossbred cows & Murrah buffaloes to raise energy density without acidosis.

Ketone bodies: acetoacetate, β-hydroxybutyrate, acetone — formed when acetyl-CoA exceeds oxaloacetate supply (negative energy balance) ⇒ **bovine ketosis**, **ovine pregnancy toxæmia**, **feline hepatic lipidosis**.

WATER — ROLE, REQUIREMENT, METABOLIC WATER

Functions (write 8): universal solvent & transport medium; medium for all hydrolytic/metabolic reactions; **thermoregulation** (high specific heat & latent heat of vaporisation); lubricant (synovial, CSF, serous); shock absorber (amniotic, aqueous humour); maintains cell turgor & shape; excretion of urea & toxins; essential for milk (**milk is 87% water**) and for ruminal fermentation.

Three sources: ① **Drinking water** ② **Feed (preformed) water** — green fodder 70–90%, straw 8–10% ③ **Metabolic water** — from oxidation of nutrients.

NUTRIENT OXIDISED (100 G)	METABOLIC WATER YIELDED
Fat	107 g
Carbohydrate	55–60 g
Protein	40–42 g

Fat gives the most metabolic water — the classic MCQ (camel hump, hibernating and desert species; kangaroo rat survives on it entirely).

Requirement (thumb rules): cattle/buffalo **3–5 L per kg DM intake** (up to 8 in summer); lactating cow adds **~3 L per L of milk**; sheep/goat **2–4 L/kg DM**; poultry **2 : 1** water : feed (doubles above 32 °C); pigs **2.5–3 L/kg feed**.

Losses: urine, faeces, evaporation (skin + lung), milk, sweat. **Water deprivation kills faster than any other nutrient deficit** — loss of **10%** body water = serious illness, **20%** = death.

Quality limits: TDS < 3000 ppm safe for cattle; nitrate-N < 100 ppm; salinity > 7000 ppm ⇒ refusal, scouring, drop in production.

MACRO-MINERALS — REQUIRED IN G/KG DIET (7)

MINERAL	CORE FUNCTION	DEFICIENCY — NAME THE DISEASE	EXCESS / INTERACTION	SOURCE & NUMBER
Calcium	99% in bone as hydroxyapatite; muscle contraction, clotting (factor IV), nerve, milk	Rickets (young) · Osteomalacia (adult) · Milk fever / parturient paresis · Cage layer fatigue , thin shells	↑Ca in dry period ⇒ milk fever; Ca ↓ Zn & P absorption	Limestone (38% Ca), DCP, bone meal; Ca:P = 2:1 (1–2:1 ruminant, 1.5:1 poultry)
Phosphorus	Bone, ATP, DNA/RNA, phospholipid, buffer; rumen microbial growth	Rickets/osteomalacia, Pica / Aphosphorosis (bone chewing) → botulism risk ; anoestrus, low yield	Excess P ⇒ urolithiasis in male sheep/goat (Ca:P <1.5:1)	Cereals/brans (as phytate P — unavailable to poultry & pigs without phytase)
Magnesium	Cofactor for >300 enzymes (all kinases/ATPases); bone; neuromuscular	Grass tetany / Hypomagnesaemic tetany in lactating cows on lush young spring pasture; milk tetany in calves	High K & N in lush grass block Mg absorption	MgO; blood Mg < 1.7 mg/dL = tetany
Sodium	Chief extracellular cation; osmotic pressure, acid–base, Na ⁺ /K ⁺ -ATPase, saliva buffer	Pica/salt hunger, ↓ appetite & growth, rough coat, ↓ milk	Excess ⇒ salt poisoning (pigs/poultry — water deprivation), nervous signs	Common salt 1–2% of concentrate ; ruminant saliva Na-bicarbonate is the rumen buffer
Potassium	Chief intracellular cation; nerve impulse, enzyme activation	Rare (forages are K-rich); anorexia, muscle weakness	Excess K antagonises Mg & raises DCAD ⇒ milk fever risk	Green fodder, molasses
Chlorine	Anion of ECF; HCl of gastric juice ; acid–base	Alkalosis, ↓ appetite, "chloride-deficiency" egg drop	—	Common salt (NaCl = 39% Na, 61% Cl)
Sulphur	In Met, Cys, biotin, thiamine, CoA, chondroitin, keratin, wool	Poor wool/hair growth, ↓ microbial protein synthesis, ↓ appetite	Excess ⇒ polioencephalomalacia (H ₂ S)	Na ₂ SO ₄ , elemental S; N:S = 10–14:1 when feeding urea

TRACE MINERALS — REQUIRED IN MG/KG (PPM). PAIR THE MINERAL WITH ITS DISEASE NAME.

ELEMENT	FUNCTION	DEFICIENCY — THE EXAM WORD	TOXICITY / ANTAGONIST
Iron	Haemoglobin, myoglobin, cytochromes, catalase	Piglet anaemia (sow milk is Fe-poor, no soil contact) — hypochromic microcytic; white scours	Prevented by iron-dextran 150–200 mg IM on day 3
Copper	Ceruloplasmin (Fe mobilisation), cytochrome oxidase, lysyl oxidase, tyrosinase, keratinisation of wool	Swayback / Enzootic ataxia in lambs · Falling disease in cattle · achromotrichia (loss of coat pigment), steely/stringy wool , anaemia	Sheep are the most Cu-susceptible species → haemolytic crisis, jaundice, "gun-metal" kidney. Antagonists: Mo, S, Fe, Zn
Cobalt	Sole role: rumen microbial synthesis of Vitamin B₁₂	Pine / Bush sickness / Coast disease — wasting, anaemia, ↑ liver ketones	Give as cobalt bullet or 1 ppm in mineral mixture
Manganese	Bone matrix (mucopolysaccharide), arginase, pyruvate carboxylase, reproduction	Perosis / slipped tendon in chicks · chondrodystrophy · silent heat & low conception in cattle	Also caused by choline, biotin, folic acid, niacin deficiency (perosis is multifactorial)
Zinc	>300 metalloenzymes (carbonic anhydrase, alkaline phosphatase); immunity, spermatogenesis, taste	Parakeratosis in swine · thymic atrophy & impaired immunity · shortened long bones, poor feathering in chicks · testicular hypoplasia	Excess dietary Ca precipitates Zn as phytate → parakeratosis on high-Ca pig diets
Iodine	Thyroxine T ₃ /T ₄ — controls BMR	Goitre ; hairless / stillborn weak calves and piglets, foetal resorption	Goitrogens : glucosinolates of Brassica/mustard cake, thiocyanates; treat with stabilised iodised salt (KIO ₃)
Selenium	Glutathione peroxidase — cytosolic antioxidant (Vit E acts in membrane)	White Muscle Disease / Nutritional Muscular Dystrophy in calves & lambs · retained placenta · exudative diathesis & pancreatic fibrosis in chicks · mulberry heart & hepatitis dietetica in pigs	Narrow safety margin: Alkali disease (chronic, hoof & hair loss) and Blind staggers (acute) from Se-accumulator plants (<i>Astragalus</i>)
Molybdenum	Xanthine oxidase, nitrate reductase (microbial)	Practically never deficient	Teart / Peat scours — Mo-induced secondary Cu deficiency (Cu–Mo–S thiomolybdate complex)
Chromium	Glucose Tolerance Factor — potentiates insulin	Impaired glucose tolerance; stress in transition cows	Cr-picolinate/yeast used as an anti-stress additive
Fluorine	Traces harden enamel	Dental caries (rare in livestock)	Fluorosis — mottled teeth, exostoses, lameness (rock phosphate, industrial & well water in Rajasthan/Gujarat/AP)

One-line disease pairs to over-learn: Se → White muscle · Cu → Swayback · Co → Pine · Mn → Perosis · Zn → Parakeratosis · I → Goitre · Fe → Piglet anaemia · Mo → Teart · Mg → Grass tetany · Ca → Milk fever · P → Pica.

FAT-SOLUBLE VITAMINS — A D E K

VIT	ACTIVE FORM / ROLE	DEFICIENCY
A Retinol	11-cis-retinal + opsin = rhodopsin ; maintains all epithelia; bone remodelling; from β-carotene	Nyctalopia (night blindness) — earliest sign; xerophthalmia , keratinisation of epithelia, pityriasis , infertility & retained placenta, ataxia from ↑ CSF pressure , nutritional roup in poultry
D Calciferol	D₃ cholecalciferol (animal, skin 7-dehydrocholesterol + UV) & D₂ ergocalciferol (sun-cured plants). Active = 1,25-(OH)₂-D₃ / calcitriol (a hormone) → ↑ Ca & P absorption	Rickets in young, Osteomalacia in adults, thin shells & cage-layer fatigue. Poultry can use only D₃ — D₂ is ~1/10 as potent (classic MCQ)
E Tocopherol	α-tocopherol — membrane-phase chain-breaking antioxidant; spares Se & PUFA	White muscle disease (with Se), Encephalomalacia / crazy chick disease , exudative diathesis , mulberry heart in pigs, testicular degeneration
K Phylloquinone	γ-carboxylation of prothrombin (II) and factors VII, IX, X . K ₁ plant, K ₂ microbial, K₃ menadione synthetic	Prolonged clotting time, haemorrhage . Sweet clover poisoning — mouldy <i>Melilotus</i> → dicoumarol antagonist; warfarin/rodenticide toxicity

Conversion: 1 IU Vit A = **0.3 µg** retinol; 1 mg β-carotene ≈ **400 IU** in cattle (poultry convert efficiently; **the cat cannot convert carotene at all** and needs preformed retinol). **Only A, D, E, K are stored** (liver) — hence they are the only vitamins that are genuinely **toxic in excess** (hypervitaminosis A & D).

Ruminant rule: adult ruminants need **no dietary B-complex or Vit C** — rumen microbes synthesise B vitamins (given adequate **cobalt**) and tissues synthesise C. The **pre-ruminant calf is functionally a monogastric** and must get dietary B vitamins.

WATER-SOLUBLE VITAMINS — B-COMPLEX & C

VITAMIN	COENZYME / ROLE	DEFICIENCY DISEASE
B₁ Thiamine	TPP — oxidative decarboxylation (pyruvate → acetyl-CoA)	Polioencephalomalacia (CCN) in ruminants from thiaminase -producing microbes/bracken fern; Chastek paralysis in fox & mink from raw fish; beriberi , opisthotonus in chicks
B₂ Riboflavin	FAD, FMN	Curled-toe paralysis in chicks (pathognomonic), poor hatchability, clubbed down
B₃ Niacin	NAD, NADP; made from tryptophan (60 mg Trp = 1 mg niacin)	Pellagra (3 D's) · Black tongue in dogs · perosis. The cat cannot convert Trp → niacin
B₅ Pantothenic acid	Coenzyme A	Goose-stepping gait in pigs (pathognomonic), dermatitis, "spectacle eye" in chicks
B₆ Pyridoxine	PLP — transamination, decarboxylation	Convulsions, microcytic anaemia, dermatitis
B₇ Biotin	CO ₂ carrier for carboxylases (pyruvate, acetyl-CoA)	Fatty Liver & Kidney Syndrome (FLKS) in broilers, foot-pad dermatitis , hoof cracks in cattle. Avidin of raw egg white binds it
B₉ Folic acid	One-carbon transfer, purine synthesis	Megaloblastic anaemia, poor feathering, perosis, anaemia in pigs
B₁₂ Cyanocobalamin	Methylmalonyl-CoA mutase — the link that lets propionate enter the TCA cycle ; contains cobalt	Anaemia, ↓ growth; in ruminants seen only as cobalt deficiency
Choline	Lipotropic factor; acetylcholine, phosphatidylcholine	Perosis & fatty liver — rumen-protected choline is fed to transition cows
C Ascorbic acid	Collagen hydroxylation (proline/lysine), antioxidant, Fe absorption	Scurvy . Dietary requirement only in guinea pig, primates, fish and some bats — all other species synthesise it from glucose

Pathognomonic quartet for MCQs: Curled-toe = B₂ · Goose-stepping = B₅ · Crazy chick = Vit E · Perosis = Mn (+choline, biotin, folic, niacin).

PARTITION OF FOOD ENERGY — DRAW THIS FLOW IN EVERY ENERGY QUESTION

GROSS ENERGY (GE)
bomb calorimeter

- faecal E
25-50% ↓

DIGESTIBLE ENERGY (DE)
= apparent

- urine 4-5%
- CH₄ 6-8% ↓

METABOLISABLE ENERGY (ME)
"available energy"

- heat increment
15-30% ↓

NET ENERGY (NE)
NE_m + NE_p

Definitions to write **verbatim**: **GE** = heat of complete combustion of a unit feed in a bomb calorimeter in excess O₂. **DE** = GE – faecal energy (*apparent*, because faeces contain metabolic faecal matter). **ME** = DE – (urinary + gaseous/methane energy). **NE** = ME – heat increment (HI); NE is split into **NE_m** (maintenance) and **NE_p** (production). **Heat increment** = **specific dynamic action**: unavoidable heat of digestion, fermentation and nutrient metabolism — **a loss in summer but useful for keeping warm in winter** (favourite viva twist).

TDN % = %DCP + %DCF + %DNFE + (%DEE × 2.25)

1 kg TDN = 4.409 Mcal DE = 4409 kcal · ME = DE × 0.82 (ruminant)

Digestibility coefficient (%) = [(Nutrient eaten - Nutrient in faeces) ÷ Nutrient eaten] × 100

Nutritive Ratio (NR) = (TDN - DCP) ÷ DCP → express as 1 : x

NR interpretation: **narrow** (1:3 to 1:5) = protein-rich, for young growing/high-yielding animals; **wide** (1:8 and above) = energy-rich, for maintenance, fattening & draft. **Calorie : protein ratio** = kcal ME per kg diet ÷ % CP — a poultry concept; broiler starter ≈ **135-140**, layer ≈ **160-175**. If it widens, birds eat to satisfy energy and become **protein-deficient and fat**.

THE ENERGY SYSTEMS — COMPARE THEM

SYSTEM	AUTHOR / COUNTRY	BASIS & THE NUMBER
TDN	USA (Henry & Morrison)	Sum of digestible nutrients, fat ×2.25. Simple but overvalues roughage (ignores methane & HI) and is a DE-level measure expressed as weight
Starch Equivalent (SE)	Kellner, Germany	kg of pure starch producing the same body fat as 100 kg of the feed. 1 kg starch = 248 g fat = 2.36 Mcal NE . Uses value factors to correct for crude fibre
Therm / NE	Armsby, USA	Net energy expressed in therms (1 therm = 1000 kcal = 1 Mcal); measured by respiration calorimeter
Scandinavian Feed Unit	Denmark/Sweden (Fjord, Hansson)	1 Feed Unit = 1 kg of barley
Physiological Fuel Value	Atwater	CHO 4 · Protein 4 · Fat 9 kcal/g (human/monogastric). Compare GE: CHO 4.1, protein 5.65, fat 9.4 — protein's ME is lower because urea is excreted still carrying energy

Kellner's fat-production values (g fat / kg digestible nutrient): starch/CHO **248** · digestible protein **235** · fat of cereals **474** · fat of oilseeds **598** · fat of roughages **474**.

Merit/demerit line: TDN is easy and needs only a digestion trial, but it is a **digestible-energy-level** value expressed in weight units, so it **overestimates roughages by 20-30%**. SE and NE account for HI and rank roughages correctly, but need a costly respiration calorimeter.

CALORIMETRY & C-N BALANCE

DIRECT calorimetry — measures **heat produced** directly. Types: **Atwater-Rosa-Benedict** adiabatic respiration calorimeter, gradient-layer calorimeter, and the **bomb calorimeter** (measures GE of the feed itself). Accurate but expensive, cumbersome, and unsuitable for large animals.

INDIRECT calorimetry — computes heat from **respiratory gas exchange** (O₂ consumed, CO₂ & CH₄ produced) and urinary N.

RQ = CO₂ produced ÷ O₂ consumed → CHO 1.00 · Protein 0.80 · Fat 0.70

Two circuits: **Closed-circuit** (Regnault-Reiset) — animal rebreathes; CO₂ absorbed by KOH, O₂ replenished and measured. **Open-circuit** — outside air drawn through, in-and-out gas concentrations analysed (used for large ruminants; today as head-hoods and SF₆ tracer). **Comparative slaughter technique** — an indirect method comparing carcass energy of a slaughtered baseline group with the fed group; **no calorimeter needed, but destructive**.

Carbon-Nitrogen balance — indirect method to know **what** was retained, not just how much: N retention ⇒ protein gain; the carbon left after accounting for protein ⇒ fat gain.

N retained = N intake - (faecal N + urinary N) · Protein gained = N retained × 6.25

C in protein = 52% · C in fat = 76.5% · N in protein = 16%

Worked logic: retained C – (protein gained × 0.52) = C stored as fat; ÷ 0.765 = **kg fat deposited**. **Positive N balance** = growth, pregnancy, lactation recovery; **negative N balance** = starvation, fever, protein deficiency, and the early lactation high-yielder.

PROTEIN EVALUATION — EVERY FORMULA, WITH WHAT EACH ONE REALLY MEASURES

Apparent Digestibility % = [(N intake - Faecal N) ÷ N intake] × 100

True Digestibility (TD) % = {[N intake - (Faecal N - Metabolic Faecal N)] ÷ N intake} × 100

Biological Value (BV) = [N retained ÷ N absorbed] × 100 = {I - (F - MFN) - (U - EUN)} ÷ {I - (F - MFN)} × 100

Net Protein Utilisation (NPU) = BV × TD ÷ 100 = N retained ÷ N intake × 100

Protein Efficiency Ratio (PER) = Body weight gain (g) ÷ Protein consumed (g)

Net Protein Ratio (NPR) = (Gain of test group + Loss of protein-free group) ÷ Protein consumed

Chemical Score = (mg limiting AA per g test protein ÷ same in whole egg) × 100

DCP % = CP % × digestibility coefficient of CP ÷ 100

How to distinguish them in one line each: **BV** = of the protein *absorbed*, how much is *retained* — a quality measure that ignores digestibility. **NPU** = BV corrected for digestibility, so it grades the protein *as fed*. **PER** = a growth-based bioassay, cheap but influenced by intake and maintenance. **NPR** corrects PER's biggest defect by adding back the maintenance loss of a protein-free control. **Chemical score** = a purely laboratory (in-vitro) index using whole egg (BV 100, the reference protein) as the standard; it ignores digestibility and availability.

Reference values: whole egg **100** · milk **85-90** · fish meal **75-80** · microbial protein **66-80** · groundnut cake **~55** · maize (zein, Lys+Trp deficient) **~50**.

RUMINANT VS NON-RUMINANT PROTEIN QUALITY

POINT	NON-RUMINANT	RUMINANT
What limits performance	Amino-acid balance of the diet	RDP:RUP balance + fermentable energy
Measure used	BV, NPU, PER, chemical score, available lysine	Degradability (in-sacco nylon bag), DCP, MP system
Can NPN be used?	No (no ureolytic microbes) — urea is toxic	Yes , up to 1% of concentrate / 3% of total DM , ≤ 1/3 of total N
Effect of heat processing	Destroys trypsin inhibitor, but Maillard reaction blocks lysine	Mild heat/formaldehyde is deliberately used to create bypass protein

Urea toxicity: excess NH₃ → rumen pH > 7.5 → free NH₃ absorbed → blood NH₃ > 1 mg/dL → tremors, bloat, tetany, death in 1-4 h. **Antidote**: **4-5 L of 5% acetic acid (vinegar) + cold water orally**. Never feed urea in drinking water, never to non-ruminants, and always adapt over **2-3 weeks**.

CLASSIFICATION OF FEEDS — THE 8 INTERNATIONAL CLASSES

#	CLASS	INDIAN EXAMPLES
1	Dry forages & roughages (>18% CF, <60% TDN)	Paddy straw, wheat bhusa, hay, sugarcane bagasse
2	Pasture, range & green forage	Berseem, lucerne, maize, sorghum/chari, hybrid Napier
3	Silages	Maize silage, sorghum silage
4	Energy feeds (<20% CP, <18% CF)	Maize, barley, broken rice, molasses, rice polish, wheat bran
5	Protein supplements (≥20% CP)	GNC, mustard cake, cottonseed cake, soybean meal, fish meal
6	Mineral supplements	Limestone, DCP, common salt, mineral mixture
7	Vitamin supplements	Vit AD ₃ E premix, riboflavin
8	Feed additives	Antibiotics, probiotics, enzymes, antioxidants, coccidiostats

Also classify by: origin (plant/animal/mineral/synthetic), CF content (concentrate <18% vs roughage >18%), moisture (dry vs succulent), and nutritive value. Legume vs non-legume fodder: legumes (berseem, lucerne, cowpea) carry higher CP (15–20%), Ca and carotene, fix atmospheric N, but cause bloat; non-legumes (maize, oats, Napier) are higher in soluble CHO and safer for silage.

FEED PROCESSING — CONCENTRATES & ROUGHAGES

Concentrates — physical: grinding (↑ surface area; too fine causes ulcers in pigs & parakeratosis of rumen), pelleting (↓ wastage & selection, ↑ density & intake, kills Salmonella), crumbling, extrusion & expansion (gelatinises starch — pet foods, aqua feed), steam flaking (↑ starch digestibility in dairy), roasting/popping/micronising, soaking & germination.

Purpose of processing (write 6): ↑ digestibility & palatability · destroy anti-nutritional factors & pathogens · ↓ selective feeding & wastage · ease storage/transport · alter rumen degradability (bypass) · permit inclusion of low-quality ingredients.

Roughage improvement — the 3-way classification:

METHOD	TECHNIQUE	EFFECT
Physical	Chaffing (2–3 cm), soaking, grinding, pelleting, steam pressure treatment, irradiation	↑ intake & ↓ wastage; little change in digestibility
Chemical	Alkali — Beckmann method (1.5% NaOH soak 24 h, then wash) & dry NaOH; Ammonia/urea treatment; Ca(OH) ₂ ; acid; oxidative (SO ₂ , H ₂ O ₂)	Breaks lignin–hemicellulose ester bonds ⇒ DM digestibility ↑ by 8–12 units
Biological	White-rot fungi (Pleurotus, Coprinus) — ligninolytic; fibrolytic enzymes (cellulase, xylanase); solid-state fermentation	Selective lignin removal; slow, but no chemical residue

Urea ammoniation of straw — memorise the numbers: 4 kg urea dissolved in 40 L water sprayed over 100 kg straw, stacked airtight (polythene/pit) and cured 21 days; aerate 24 h before feeding. Result: CP rises from 3–4% to 8–10%, DM digestibility ↑ ~10 units, intake ↑ 25–30%. Cheapest single intervention for the Indian straw-based diet.

CONSERVATION — SILAGE VS HAY

SILAGE = fodder preserved by controlled anaerobic lactic-acid fermentation at high moisture. Ensiling proceeds in 4 phases: ① Aerobic (plant respiration uses residual O₂, temp ↑ 25–35 °C) ② Fermentation (Lactobacillus, Streptococcus, Pediococcus convert sugars → lactic acid, pH falls) ③ Stable/storage (pH < 4.2, all activity ceases) ④ Feed-out/aerobic spoilage (yeasts & moulds on air exposure).

QUALITY INDEX	GOOD SILAGE
Crop DM at ensiling	30–35%
pH	3.8–4.2
Lactic acid	3–13% of DM (dominant)
Butyric acid	< 0.2%
NH ₃ -N (as % of total N)	< 10%
Colour / smell	Greenish-yellow to olive-brown; pleasant acidic-fruity

Ideal crops: high soluble-CHO, low buffering — maize, sorghum, oats, hybrid Napier, sugarcane tops. Legumes ensile poorly (high protein buffering) — wilt them or add molasses. Additives: molasses 1–3%, urea 0.5–1%, salt, formic acid, Lactobacillus inoculants, cellulase. Silos: trench, pit, bunker, tower, and baled/wrapped silage. Butyric (clostridial) silage = rancid, pH > 5, NH₃-N high — caused by wet, soil-contaminated, poorly compacted material. Flieg's score grades silage from the relative proportions of lactic : acetic : butyric acid.

HAY = fodder preserved by drying to <15–20% moisture so enzymes and microbes cannot act. Cut at 50% flowering — the compromise point between yield and quality. Methods: field/swath curing, tripod & barn drying, artificial dehydration. Best hay crops: lucerne, cowpea, oats, Cenchrus, Stylosanthes. Losses: shattering of leaves (the leaf holds most protein & carotene), bleaching (carotene destroyed by sunlight), leaching by rain, and spontaneous heating → fire risk if baled too wet.

One-line comparison: Hay = dry preservation, weather-dependent, carotene lost, cheap storage. Silage = wet preservation, weather-independent, carotene & nutrients retained, needs airtight structure and heavy initial cost.

HARMFUL NATURAL CONSTITUENTS (ANFS) & ADULTERANTS

TOXIN	SOURCE	LESION / SIGN & REMEDY
HCN (dhurrin)	Young/drought-stressed sorghum & chari < 50 days, Sudan grass, cassava	Cytochrome oxidase block; bright cherry-red blood , sudden death. Antidote: sodium nitrite + sodium thiosulphate IV . Never graze sorghum below 3 ft / 50 days
Gossypol	Raw cottonseed cake (free gossypol)	Cardiac & hepatic damage, ↓ fertility in bulls; monogastrics susceptible. Limit < 0.04% free gossypol ; treat with FeSO₄ 1:1 (Fe:gossypol) or heat
Mimosine / DHP	Subabul (<i>Leucaena leucocephala</i>)	Alopecia, salivation, goitre; limit to < 30% of ration ; <i>Synergistes jonesii</i> degrades DHP
Oxalates	Hybrid Napier, setaria, <i>Amaranthus</i> , sugarbeet tops	Bind Ca ⇒ hypocalcaemia; "Big head" (osteodystrophia fibrosa) in horses ; renal oxalate crystals
Nitrate → Nitrite	Heavily N-manured lush oats, maize, weeds; polluted water	Methaemoglobinaemia — chocolate-brown blood , dyspnoea. Antidote: 1% methylene blue IV @ 4–15 mg/kg
Glucosinolates / goitrogens	Mustard & rapeseed cake, Brassica	Goitre, ↓ growth; limit mustard cake to 20–25%; iodine supplementation
Trypsin inhibitor, haemagglutinin	Raw soybean	Pancreatic hypertrophy, poor growth. Destroyed by moist heat 110 °C for 15–20 min ; verified by the urease activity test
Thiaminase	Bracken fern , raw fish	Polioencephalomalacia; Chastek paralysis in fur animals
Aflatoxin B ₁	<i>Aspergillus flavus</i> on damp groundnut cake & maize (>14% moisture)	Hepatotoxic, carcinogenic, immunosuppressive; M₁ excreted in milk . Limit 20 ppb ; use toxin binders (HSCAS, esterified glucomannan)
BOAA / ODAP	Lathyrus sativus (khesari dal)	Neurolathyrism — spastic paraplegia; steeping & parboiling reduce it
Tannins, saponins, phytate	Sorghum, tea waste, mahua cake, cereal brans	Bind protein/minerals; phytate P is unavailable to poultry & pigs ⇒ add phytase

Common adulterants & their detection: sand/silica in brans (⇒ **AIA > 3%**) · urea in oil cakes (urease/Nessler’s test) · **marble/limestone powder** in mineral mixture (effervescence with HCl) · argemone seed in mustard cake (nitric acid test) · **sawdust & husk** in wheat bran (floatation) · **mahua & kapok cake** in GNC (microscopy) · **common salt** excess (AgNO₃) · **melamine** as a false-protein adulterant (Kjeldahl reads N, not true protein — *the reason NPN cannot be detected by CP alone*).

FEED ADDITIVES & SUPPLEMENTS — NON-NUTRITIVE BUT MARKS-HEAVY

CLASS	EXAMPLES	MODE OF ACTION / EXAM POINT
Antibiotic growth promoters	Bacitracin, virginiamycin, tylosin	Suppress subclinical gut infection, thin gut wall ⇒ better absorption. Now restricted/banned in India & EU over AMR & residues
Ionophores	Monensin , lasalocid, salinomycin	Shift rumen VFA toward propionate , ↓ methane, ↓ acidosis & bloat, coccidiostatic. FATAL to horses
Probiotics (DFM)	<i>Saccharomyces cerevisiae</i> , <i>Lactobacillus</i> , <i>Bacillus subtilis</i>	Live microbes — competitive exclusion, stabilise rumen pH, ↑ fibre digestion
Prebiotics	MOS, FOS, inulin	Non-digestible substrates that selectively feed beneficial flora / bind pathogen fimbriae
Enzymes	Phytase , xylanase, β-glucanase, cellulase	Phytase releases phytate-P ⇒ less inorganic P needed and less P pollution ; NSPase cuts gut viscosity in wheat/barley broiler diets
Buffers	Sodium bicarbonate 0.75–1.5% of DM , magnesium oxide	Stabilise rumen pH on high-grain diets ⇒ prevent SARA and milk-fat depression
Antioxidants	Ethoxyquin, BHT, BHA	Prevent rancidity of fats and destruction of vitamins A & E in stored feed
Mould inhibitors / binders	Propionic acid, sodium propionate; HSCAS, bentonite, esterified glucomannan	Prevent fungal growth; adsorb mycotoxins in the gut
Hormone-like & misc.	Pellet binders (bentonite), flavours, pigments (xanthophyll), coccidiostats, methane inhibitors	Hormonal growth promoters are banned in food animals in India

Additive vs Supplement — the definition question: a **supplement** supplies a **nutrient** that the basal ration lacks (protein, mineral, vitamin) and is fed in appreciable quantity; an **additive** is a **non-nutritive** substance included in very small amounts to improve feed quality, efficiency of utilisation, animal health or product quality.

LAST-10-MINUTES DRILL — ONE-LINE FACTS MOST OFTEN ASKED

Father of nutrition — **Lavoisier**
Proximate analysis year/place — **1860, Weende, Germany**
Fraction obtained by difference — **NFE**
Ash temperature — **550–600 °C**
Sand adulteration indicator — **Acid Insoluble Ash**
CP factor — **N × 6.25**
Fat energy multiplier in TDN — **2.25**
1 kg TDN = — **4.409 Mcal DE**
Predicts intake / digestibility — **NDF / ADF**
Only glucogenic VFA — **Propionate**
Milk-fat precursor VFA — **Acetate**
Normal acetate:propionate — **3 : 1**
Rumen fuel of epithelium — **Butyrate**
Methane loss of GE — **6–8%**
Most metabolic water — **Fat (107 g/100 g)**
Reference protein (BV 100) — **Whole egg**
NPU = — **BV × TD / 100**
Protein equivalent of urea — **281–291%**
Max urea in concentrate — **1%**
Urea toxicity antidote — **5% acetic acid + cold water**
N : S ratio for NPN — **10–14 : 1**
Starch Equivalent author — **Kellner**
Net Energy/Therm author — **Armsby**
1 kg starch = NE — **2.36 Mcal (248 g fat)**
Scandinavian Feed Unit — **1 kg barley**
Atwater values — **CHO 4 · Pro 4 · Fat 9**
RQ of fat / CHO — **0.70 / 1.00**
Carbon in protein / fat — **52% / 76.5%**
Concentrate vs roughage CF — **<18% vs >18%**
Ideal Ca:P — **2 : 1**
Ideal silage pH — **3.8–4.2**
Silage DM at ensiling — **30–35%**
Hay moisture — **<15–20%**
Hay cutting stage — **50% flowering**
Urea-treatment recipe — **4 kg urea + 40 L water / 100 kg straw, 21 d**
Beckmann method — **1.5% NaOH, 24 h**
Aflatoxin limit / organism — **20 ppb · A. flavus**
Cherry-red blood — **HCN poisoning**
Chocolate-brown blood — **Nitrite (methaemoglobin)**
Raw soybean factor — **Trypsin inhibitor**
Test for soybean heating — **Urease activity**
Ionophore fatal to — **Horses (monensin)**
Rumen buffer dose — **NaHCO₃ 0.75–1.5% DM**
Enzyme freeing phytate-P — **Phytase**
Vitamin poultry cannot use — **D₂ (needs D₃)**
Vitamin not needed by adult ruminant — **B-complex & C**
Species needing dietary Vit C — **Guinea pig, primates, fish**
Cat-only essentials — **Taurine, arachidonic acid, preformed Vit A, niacin**
Narrow vs wide NR — **1:4 growth · 1:8+ fattening/draft**